

CYBERSECURITY SCANNING PROJECT REPORT

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PROJECT TITLE : ADVANCED STEALTH SCANNING AND IDS
EVASION TECHNIQUES IN A STIMULATED ENTERPRISE NETWORK

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Executive Summary

This project involved simulating a covert reconnaissance attack against a segmented enterprise network protected by a SNORT IDS. Various Nmap, masscan and Hping3 stealth scanning techniques were employed to assess the IDS's detection capabilities to identify open ports and services on the target system without triggering the IDS alerts. The report details the methods that successfully evaded detection and those that triggered alerts, concluding with recommendations for enhancing IDS rule sets.

Note: Testing was conducted on a personally owned, isolated lab network.

Lab Environment & Methodology

Attacker Machine: Kali Linux [IP: 10.10.10.10]

Tools -- Running Nmap, Hping3, and masscan..

Target Machine: Metasploitable [IP: 10.10.10.20]

-- Vulnerable machine..

IDS Sensor: Ubuntu [IP: 10.10.10.30]

Tools -- Running SNORT IDS in promiscuous mode and Wireshark.

For each test we document: objective → command(s) used → expected detection behavior → actual results (alerts, pcap snippet). Keep tests atomic (one evasion technique per run).

SCANNING TECHNIQUES AND RESULTS

1) TCP SYN scan — commonly called a SYN scan

SCAN TECHNIQUE	COMMMAND USED	TOOLS
TCP SYN Stealth Scan	Nmap -sS <IP addr>	Nmap

FINDINGS : An Nmap SYN scan (-sS) checks ports by sending a SYN packet (the first step of a TCP connection). If the target replies with SYN/ACK, the port is open but If

there's no reply, the port is filtered (likely blocked by a firewall). it just tells you whether ports are open, closed, or filtered...

```
(venv)-(root@kali)-[/home/cyberxdiety/Desktop]
# nmap -sS 10.10.10.20
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-18 18:02 EDT
map scan report for 10.10.10.20
Host is up (0.00045s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
1/tcp     open  ftp
2/tcp     open  ssh
3/tcp     open  telnet
5/tcp     open  smtp
3/tcp     open  domain
80/tcp    open  http
11/tcp    open  rpcbind
39/tcp    open  netbios-ssn
45/tcp    open  microsoft-ds
12/tcp    open  exec
13/tcp    open  login
14/tcp    open  shell
899/tcp   open  rmiregistry
524/tcp   open  ingreslock
```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {YES}

```
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Commencing packet processing (pid=1919)
9/18-18:02:31.948522  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.10:36500 -> 10.10.10.20:139
9/18-18:02:31.948522  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.10:36500 -> 10.10.10.20:139
9/18-18:02:31.949378  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.20:139 -> 10.10.10.10:36500
9/18-18:02:31.949378  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.20:139 -> 10.10.10.10:36500
9/18-18:02:31.949386  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.10:36500 -> 10.10.10.20:110
9/18-18:02:31.949386  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.10:36500 -> 10.10.10.20:110
9/18-18:02:31.950243  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.20:110 -> 10.10.10.10:36500
9/18-18:02:31.950243  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.20:110 -> 10.10.10.10:36500
```

Conclusion- Detection of Nmap Scanning by Snort

During this reconnaissance phase I executed three common Nmap scan types against the lab hosts:

```
nmap -sS <ip addr > (SYN)
nmap -sT <ip addr > (TCP connect),
nmap -sU <ip addr > (UDP).
```

All three techniques were successfully detected by the Snort IDS. Snort generated corresponding alerts during each scan run, and these alerts correlate with packet captures taken at the time of testing. This result demonstrates that, in this environment and with the active ruleset/preprocessors, Snort is effective at identifying basic TCP and UDP reconnaissance activity. It indicates that an attacker using these common scanning methods would likely be noticed, allowing defenders to escalate, investigate, and respond.

2) FIN/NULL/XMAS SCAN

SCAN TECHNIQUE	COMMMAND USED	TOOLS
Fin scan technique	nmap -sF <IP addr>	Nmap / snort IDS
Null scan technique	Nmap -sN <ip addr>	Nmap / snort IDS

Xmas scan technique	Nmap -sX <ip addr>	Nmap / snort IDS

FINDINGS : NULL, FIN and XMAS scans are *reconnaissance* techniques used to map TCP ports while trying to be stealthier than a classic SYN scan. The goal is to discover which ports are open or closed by exploiting differences in how TCP stacks respond to unexpected flag combinations — ideally leaving less obvious footprints than a full handshake.

```

└─$ nmap -sN 10.10.10.20
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-19 06:31 EDT
Nmap scan report for 10.10.10.20
Host is up (0.00022s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
23/tcp    open|filtered telnet
25/tcp    open|filtered smtp
53/tcp    open|filtered domain
80/tcp    open|filtered http
111/tcp   open|filtered rpcbind
139/tcp   open|filtered netbios-ssn

```

```

(venv)~(root@kali)-[/home/cyberxdiety/Desktop]
└─$ nmap -sX 10.10.10.20
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-19 06:30 EDT
Nmap scan report for 10.10.10.20
Host is up (0.00040s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
23/tcp    open|filtered telnet
25/tcp    open|filtered smtp
53/tcp    open|filtered domain
80/tcp    open|filtered http
111/tcp   open|filtered rpcbind
139/tcp   open|filtered netbios-ssn
445/tcp   open|filtered microsoft-ds
512/tcp   open|filtered exec
513/tcp   open|filtered login

```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {YES}

```
09/19-06:31:36.926391  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
] {TCP} 10.10.10.10:52444 -> 10.10.10.20:6567
09/19-06:31:36.926391  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
] {TCP} 10.10.10.10:52444 -> 10.10.10.20:6567
09/19-06:31:36.926391 10.10.10.10:52444 -> 10.10.10.20:6567
TCP TTL:49 TOS:0x0 ID:12744 IpLen:20 DgmLen:40
***** Seq: 0x41507090 Ack: 0x0 Win: 0x400 TcpLen: 20
-----
```

```
09/19-06:31:37.988665  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
] {TCP} 10.10.10.10:52444 -> 10.10.10.20:10024
09/19-06:31:37.988665  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
] {TCP} 10.10.10.10:52444 -> 10.10.10.20:10024
09/19-06:31:37.988665 10.10.10.10:52444 -> 10.10.10.20:10024
TCP TTL:43 TOS:0x0 ID:27110 IpLen:20 DgmLen:40
***** Seq: 0x41507090 Ack: 0x0 Win: 0x400 TcpLen: 20
-----
```

Conclusion- Detection of Nmap Scanning by Snort

During the testing I executed a set of TCP-flag stealth scans (NULL, FIN, and XMAS) against the target host using `nmap -sN`, `nmap -sF`, and `nmap -sX`. Those techniques are used to map TCP ports while trying to be stealthier than a classic SYN scan. The goal is to discover which ports are open or closed by exploiting differences in how TCP stacks respond to unexpected flag combinations — ideally leaving less obvious footprints than a full handshake. But surprisingly the lab Snort sensor detected and logged each of these scan attempts. This indicates the IDS is configured to catch TCP-flag based reconnaissance patterns and that the tested signature

ruleset successfully identified the probe traffic.. Because these scans were detected, the techniques did not achieve stealth against the deployed Snort configuration in this simulated enterprise environment.

3) FRAGMENTED SYN SCAN WITH DECOY

SCAN TECHNIQUE	COMMMAND USED	ToolS
Syn scan with decoys	Nmap -sS -f -D RND:10 -oA fragmented_decoy <target ip addr>	Nmap tcpdump

FINDINGS:: The command “ nmap -sS -f -D RND:10 -oA fragmented_decoy <ip addr> is an nmap command designed to perform a stealthy TCP SYN scan on a target IP address <10.10.10.30> with packet fragmentation and decoy IP addresses to obscure the scan's origin.

```
(venv)-(root@kali)-[/home/cyberxdiety/Desktop]
# nmap -sS -f -D RND:10 -oA fragmented_decoy 10.10.10.30
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 06:52 EDT
Nmap scan report for 10.10.10.30
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
543/tcp   open  login
```


IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {NO}

Successful Source IP Evasion During Fragmented Decoy Scan

```
cyberxdiety@cyberxdiety-VirtualBox:~/Desktop$ sudo tcpdump -i enp0s3
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
11:48:50.250337 ARP, Request who-has 10.10.10.30 tell 10.10.10.10, length 46
11:48:50.250630 ARP, Reply 10.10.10.30 is-at 08:00:27:13:82:65 (oui Unknown), length 46
11:49:03.378356 IP 161.229.171.215.42974 > 10.10.10.30.netbios-ssn: [!tcp]
11:49:03.378357 IP 161.229.171.215 > 10.10.10.30: tcp
11:49:03.378357 IP 161.229.171.215 > 10.10.10.30: tcp
11:49:03.378357 IP 3.211.159.206.42974 > 10.10.10.30.netbios-ssn: [!tcp]
11:49:03.378357 IP 3.211.159.206 > 10.10.10.30: tcp
11:49:03.378357 IP 3.211.159.206 > 10.10.10.30: tcp
11:49:03.378357 IP 110.67.64.122.42974 > 10.10.10.30.netbios-ssn: [!tcp]
11:49:03.378596 IP 110.67.64.122 > 10.10.10.30: tcp
11:49:03.378596 IP 110.67.64.122 > 10.10.10.30: tcp
11:49:03.378596 IP 43.154.164.27.42974 > 10.10.10.30.netbios-ssn: [!tcp]
11:49:03.378820 IP 43.154.164.27 > 10.10.10.30: tcp
11:49:03.378820 IP 43.154.164.27 > 10.10.10.30: tcp
11:49:03.378820 IP 181.143.87.91.42974 > 10.10.10.30.netbios-ssn: [!tcp]
11:49:03.379107 IP 181.143.87.91 > 10.10.10.30: tcp
11:49:03.379107 IP 181.143.87.91 > 10.10.10.30: tcp
11:49:03.379107 IP 144.49.63.212.42974 > 10.10.10.30.netbios-ssn: [!tcp]
11:49:03.379494 IP 144.49.63.212 > 10.10.10.30: tcp
11:49:03.379494 IP 144.49.63.212 > 10.10.10.30: tcp
11:49:03.379494 IP 53.237.91.160.42974 > 10.10.10.30.netbios-ssn: [!tcp]
```

Conclusion- Detection of Scanning by Snort

Successful Source IP Evasion During Fragmented Decoy Scan

After the scan, the tcpdump capture file showed that the tool/IDS couldn't identify the scanner's true IP address. The file included broken-up TCP SYN packets from 10 different fake IP addresses (decoys), with no clear

connection to the actual sender. In a real-world attack, this could allow adversaries to map open ports and services without triggering alerts.

4) SYN stealth scan that combines packet fragmentation, payload padding, and moderate timing

SCAN TECHNIQUE	COMMAND USED	Tools	
SYN stealth scan + fragmentation	<code>nmap -sS -f --data-length 10 --mtu 24 -T3 -v <target></code>	Nmap snort IDS	

FINDINGS : `nmap -sS -f --data-length 10 --mtu 24 -T3 -v <target>`” is a SYN stealth scan that combines packet fragmentation, payload padding, and moderate timing. In plain terms: Nmap sends SYNs (half-open connection attempts) but splits the packets into small fragments and adds 10 bytes of padding to each probe — all while running at a moderately fast timing profile and showing verbose output.

```
(venv)-(root@kali)-[/home/cyberxdiety/Desktop]
# nmap -sS -f --data-length 10 --mtu 24 -T3 -v 10.10.10.20
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-19 18:50 EDT
Initiating ARP Ping Scan at 18:50
Scanning 10.10.10.20 [1 port]
Completed ARP Ping Scan at 18:50, 0.22s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 18:50
Completed Parallel DNS resolution of 1 host. at 18:50, 13.01s elapsed
Initiating SYN Stealth Scan at 18:50
Scanning 10.10.10.20 [1000 ports]
Discovered open port 111/tcp on 10.10.10.20
Discovered open port 5900/tcp on 10.10.10.20
Discovered open port 21/tcp on 10.10.10.20
Discovered open port 53/tcp on 10.10.10.20
Discovered open port 25/tcp on 10.10.10.20
Discovered open port 445/tcp on 10.10.10.20
Discovered open port 22/tcp on 10.10.10.20
Discovered open port 23/tcp on 10.10.10.20
Discovered open port 139/tcp on 10.10.10.20
```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {YES}

```
09/19-11:57:01.659690  [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.20:28201 -> 10.10.10.10:57866
09/19-11:57:01.659690  10.10.10.20:28201 -> 10.10.10.10:57866
TCP TTL:64 TOS:0x0 ID:0 IpLen:20 DgmLen:40 DF
**A*R** Seq: 0x0 Ack: 0x195E71C2 Win: 0x0 TcpLen: 20
+=====+
09/19-11:57:01.659690  [**] [1:2052:4] IP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.10 -> 10.10.10.20
09/19-11:57:01.659690  10.10.10.10 -> 10.10.10.20
TCP TTL:53 TOS:0x0 ID:35582 IpLen:20 DgmLen:44 MF
Frag Offset: 0x0000 Frag Size: 0x0018
+=====+
```

Conclusion- Detection of Nmap Scanning by Snort

The SYN scan using fragmentation, MTU forcing and payload padding (-sS -f --data-length 10 --mtu 24) was detected by the IDS. This shows the IDS' packet reassembly and signature/preprocessor logic are effective against basic fragmentation and padding

evasion techniques. However, detection alone doesn't guarantee full coverage — tune and validate reassembly settings, ensure relevant preprocessors (IP defragmentation, TCP stream reassembly) are enabled, and review rule thresholds to reduce false negatives for more sophisticated evasions.

5) HOST DISCOVERY-- commonly called Ping scan or Ping sweep

SCAN TECHNIQUE	COMMMAND USED	Tools
Ping scan technique	Nmap -sn <ip addr>	Nmap

FINDINGS:: nmap -sn <IP address> performs a host discovery (ping scan) to check if a target is online without scanning its ports. It uses ARP on local networks or ICMP/TCP probes on remote networks, and simply reports whether the host is up.

```
(venv)-(root@kali)-[/home/cyberxdiet/Desktp]
# nmap -sn 10.10.10.30
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-27 11:15 EDT
Nmap scan report for 10.10.10.30
Host is up (0.0026s latency).
MAC Address: 08:00:27:13:82:65 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 13.25 seconds
```

IDS RESPONSE :

Alert triggered {NO}

SCAN attempted and detected {NO}

Commencing packet processing (pid=2027)

Conclusion- Detection of Nmap Scanning by Snort

When running `nmap -sn <target>`, Snort IDS did not generate alerts. This is because `-sn` performs only host discovery (via ARP/ICMP/TCP probes) without full port scanning. On a local network, Nmap defaults to ARP requests, which Snort does not normally inspect unless configured. Additionally, a single ICMP echo or SYN probe often does not meet Snort's detection thresholds, so such scans may evade basic IDS detection."

6) Nmap FRAGMENTATION (MTU) scan

SCAN TECHNIQUE	COMMMAND USED	Tools
Fragmentation technique	Nmap -mtu 16 <ip addr>	Nmap

FINDINGS:: nmap --mtu 16 <IP> forces Nmap to break its probe packets into *very small IP fragments* (16- byte fragments) before sending them to the target. It's an evasion tweak which reveal port of the target — not a different scan type.

```
(venv)-(root@kali)-[/home/cyberxdieety/Desktop]
# nmap --mtu 16 10.10.10.30
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-27 11:46 EDT
Nmap scan report for 10.10.10.30
Host is up (0.00068s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {YES}


```
09/27-12:12:33.456292 0x0 [**] [1:2052:4] IP scan attempted adn Detected [**] [Priority: 0]
{TCP} 10.10.10.30:56737 -> 10.10.10.10:41877
09/27-12:12:33.456292 0x0 [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.30:56737 -> 10.10.10.10:41877
09/27-12:12:33.456292 10.10.10.30:56737 -> 10.10.10.10:41877
TCP TTL:64 TOS:0x0 ID:0 IpLen:20 DgmLen:40 DF
***A*R** Seq: 0x0 Ack: 0x43CE6DC4 Win: 0x0 TcpLen: 20
+++++
09/27-12:12:33.456455 0x0 [**] [1:2052:4] IP scan attempted adn Detected [**] [Priority: 0]
{TCP} 10.10.10.30:2161 -> 10.10.10.10:41877
09/27-12:12:33.456455 0x0 [**] [1:513:1] TCP scan attempted and Detected [**] [Priority: 0]
{TCP} 10.10.10.30:2161 -> 10.10.10.10:41877
09/27-12:12:33.456455 10.10.10.30:2161 -> 10.10.10.10:41877
TCP TTL:64 TOS:0x0 ID:0 IpLen:20 DgmLen:40 DF
***A*R** Seq: 0x0 Ack: 0x43CE6DC4 Win: 0x0 TcpLen: 20
+++++
```

Conclusion- Detection of Nmap Scanning by Snort

After i executed `nmap --mtu 16 <IP>` to test fragmentation-based evasion. Snort IDS detected the activity, indicating the IDS either reassembled fragments or matched fragmentation/anomaly signatures. Fragmentation alone is not a reliable evasion vector against the current detection stack.

7) DECOY SCAN {Nmap}

SCAN TECHNIQUE	COMMMAND USED	Tools
decoy-based IP spoofing	Nmap -sS -D RND:20 <ip addr>	Nmap snort

Decoy scan	Nmap -D RND:20 <ip addr>	Nmap tcpdump
------------	-----------------------------	-----------------

FINDINGS::

```

(venv)-(root@kali)-[/home/cyberxdiety/Desktop]
# nmap -D RND:20 10.10.10.30
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 12:38 EDT
Nmap scan report for 10.10.10.30
Host is up (0.00077s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
143/tcp   open  microsoft-ds
162/tcp   open  exec
2202/tcp  open  login

```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {NO}

```

2:38:26.730755 ARP, Reply 10.10.10.30 is-at 08:00:27:13:82:65 (oui Unknown), length 46
2:38:39.839362 IP 3.20.129.251.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606,
win 1024, options [mss 1460], length 0
2:38:39.839362 IP 203.136.28.67.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606
win 1024, options [mss 1460], length 0
2:38:39.839362 IP 218.241.13.150.59086 > 10.10.10.30.http-alt: Flags [S], seq 18829760
, win 1024, options [mss 1460], length 0
2:38:39.839362 IP 221.179.126.134.59086 > 10.10.10.30.http-alt: Flags [S], seq 1882976
6, win 1024, options [mss 1460], length 0
2:38:39.839362 IP 61.208.125.67.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606
win 1024, options [mss 1460], length 0
2:38:39.839362 IP 46.22.56.47.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606,
in 1024, options [mss 1460], length 0
2:38:39.839362 IP 215.59.102.120.59086 > 10.10.10.30.http-alt: Flags [S], seq 18829760
, win 1024, options [mss 1460], length 0
2:38:39.839363 IP 124.2.189.196.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606
win 1024, options [mss 1460], length 0
2:38:39.839521 IP 10.10.10.10.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606,
in 1024, options [mss 1460], length 0
2:38:39.839521 IP 166.54.234.96.59086 > 10.10.10.30.http-alt: Flags [S], seq 188297606
win 1024, options [mss 1460], length 0
2:38:39.839521 IP 171.144.38.192.59086 > 10.10.10.30.http-alt: Flags [S], seq 18829760
, win 1024, options [mss 1460], length 0

```

Conclusion- Detection of Scanning by Snort

Running nmap -D RND:20 <target> injected 20 randomized decoy source IPs alongside real probes; the scan returned expected port results locally while the target's logs showed probes from many different source IPs, demonstrating noise-based attribution obfuscation.

8) SOURCE IP/PORT SPOOFING

SCAN TECHNIQUE	COMMMAND USED	Tools
Port Spoofing	Hping3 -S -p 80 -a	Hping3

	<spoof ip> <target Ip>	
Port Spoofing	hping3 -S -p 80 --rand-source <ip addr>	Hping3

FINDINGS:: hping3 -S -p 80 -a <spoof ip> <target Ip> utilizes the hping3 tool to send custom TCP/IP packets, specifically designed for network testing and security assessments. this command is used to simulate TCP SYN packets directed at any port of the target IP, with the source IP address spoofed to appear as <spoof ip>

```

—(venv)—(root@kali)—[/home/cyberxdiety/Desktop]
—# hping3 -S -p 80 -a 10.10.10.40 10.10.10.30
PING 10.10.10.30 (eth0 10.10.10.30): S set, 40 headers + 0 data bytes
C
— 10.10.10.30 hping statistic —
6 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {NO}

```
11:57:06.953476 IP 10.10.10.40.2322 > 10.10.10.30.http: Flags [S], seq 1370187249, win 512, length 0
11:57:07.953962 IP 10.10.10.40.2323 > 10.10.10.30.http: Flags [S], seq 1502059925, win 512, length 0
11:57:07.960655 ARP, Request who-has 10.10.10.40 tell 10.10.10.30, length 46
11:57:08.954175 IP 10.10.10.40.2324 > 10.10.10.30.http: Flags [S], seq 247533031, win 512, length 0
11:57:08.960638 ARP, Request who-has 10.10.10.40 tell 10.10.10.30, length 46
11:57:09.954267 IP 10.10.10.40.2325 > 10.10.10.30.http: Flags [S], seq 552598443, win 512, length 0
```

This command is particularly relevant for IDS evasion testing due to its IP spoofing capability. By forging the source IP address (-a <spoof ip>) ..

Conclusion- Detection of Scanning by Snort

The command “hping3 -S -p 80 -a <spoof ip> <target Ip> “ was run on a system with appropriate permissions, sending TCP SYN packets to port 80 of the target IP <10.10.10.30>, with the source IP spoofed to a chosen address <10.10.10.40> .. Upon executing the command, the IDS logged the incoming TCP SYN packets but attributed them to the spoofed IP address (<10.10.10.40) rather than the actual source IP of the machine running the command <10.10.10.10>.

9) SYN flood attack

SCAN TECHNIQUE	COMMMAND USED	Tools
Syn flood attack	hping3 -S -p 80 --rand-source --flood	Nmap tcpdump

	<target ip>	
--	-------------	--

FINDINGS:: The command `hping3 -S -p 80 --rand-source --flood <target ip>` is a classic example of a TCP SYN flood attack with IP spoofing, specifically tuned for IDS evasion through traffic obfuscation and volume overload. It's not a stealthy "scan" but an aggressive DoS technique that disrupts services while challenging detection systems.

```

—(venv)—(root@kali)—[/home/cyberxdiety/Desktop]
# hping3 -S -p 80 --rand-source --flood 10.10.10.30
HPING 10.10.10.30 (eth0 10.10.10.30): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
31: ^C
— 10.10.10.30 hping statistic —
241151 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {NO}


```
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
12:04:52.833488 IP 170.245.81.97.1200 > 10.10.10.30.http: Flags [S], seq 2096096364, win 512, length 0
12:04:52.833489 IP 59.194.100.108.1201 > 10.10.10.30.http: Flags [S], seq 1769705234, win 512, length 0
12:04:52.833489 IP 221.20.4.123.1202 > 10.10.10.30.http: Flags [S], seq 566305581, win 512, length 0
12:04:52.833489 IP 35.27.243.216.1203 > 10.10.10.30.http: Flags [S], seq 674609286, win 512, length 0
12:04:52.833489 IP 143.110.83.216.1204 > 10.10.10.30.http: Flags [S], seq 1137776031, win 512, length 0
12:04:52.833489 IP 196.142.16.198.1205 > 10.10.10.30.http: Flags [S], seq 498182936, win 512, length 0
12:04:52.833489 IP 155.171.94.75.1206 > 10.10.10.30.http: Flags [S], seq 838727706, win 512, length 0
12:04:52.833795 IP 10.83.13.97.1207 > 10.10.10.30.http: Flags [S], seq 1639529586, win 512, length 0
12:04:52.833795 IP 198.143.20.87.1208 > 10.10.10.30.http: Flags [S], seq 1976807497, win 512, length 0
12:04:52.833795 IP 235.108.70.198.1209 > 10.10.10.30.http: Flags [S], seq 1938762702, win 512, length 0
```

Randomized IPs can blend the malicious traffic with legitimate traffic, making it harder for Snort to distinguish between normal and suspicious activity..

Conclusion- Detection of Scanning by Snort IDS

The test demonstrated that the use of source IP randomization in conjunction with SYN flooding significantly reduced the likelihood of detection by signature-based IDS systems like Snort. This highlights a potential limitation in the current IDS configuration, particularly in detecting traffic with spoofed or frequently changing source IPs. To improve detection capabilities, it is recommended to implement enhanced Snort rules tailored to identify SYN flood patterns with spoofed IPs, complemented by behavioral analysis tools for anomaly detection. Regular testing and tuning of IDS rules are essential to ensure robust defense against evasion techniques.

10) Nmap decoy scan.

SCAN TECHNIQUE	COMMMAND USED	Tools
Decoy scan	<code>nmap -D <decoy ip> <decoy ip> <Me"lp> <target lp></code>	Nmap tcpdump

FINDINGS : The “-D” option enables Nmap's decoy feature, which allows me to specify a list of IP addresses (decoys) to mask the origin of the scan. By including in the list, i instruct Nmap to include my own IP address among the decoys. This makes it appear as though the scan is coming from multiple IP addresses, potentially confusing the target's intrusion detection systems and making it harder to trace the scan back to you. The is the host or network being scanned.

```
(venv)-(root@kali)-[/home/cyberxdiety/Desktop]
└─$ nmap -D 10.10.10.50,10.10.10.60,10.10.10.80,10.10.10.90,ME 10.10.10.30
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 13:37 EDT
Nmap scan report for 10.10.10.30
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
```

IDS RESPONSE :

Alert triggered {YES}

TCP SCAN attempted and detected {NO}

```
13:37:32.248406 IP 10.10.10.60.38125 > 10.10.10.30.mysql: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.248407 IP 10.10.10.80.38125 > 10.10.10.30.mysql: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.249755 IP 10.10.10.90.38125 > 10.10.10.30.mysql: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.249756 IP 10.10.10.10.38125 > 10.10.10.30.mysql: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.249756 IP 10.10.10.50.38125 > 10.10.10.30.5900: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.249757 ARP, Request who-has 10.10.10.50 tell 10.10.10.30, length 46
13:37:32.249757 IP 10.10.10.30.mysql > 10.10.10.10.38125: Flags [S.], seq 141832635, ack 1933245247, win 5840, options [mss 1460], length 0
13:37:32.251090 IP 10.10.10.10.38125 > 10.10.10.30.mysql: Flags [R], seq 1933245247, win 0, length 0
13:37:32.256142 IP 10.10.10.60.38125 > 10.10.10.30.5900: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.257330 IP 10.10.10.80.38125 > 10.10.10.30.5900: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
13:37:32.259804 ARP, Request who-has 10.10.10.60 tell 10.10.10.30, length 46
13:37:32.259805 ARP, Request who-has 10.10.10.80 tell 10.10.10.30, length 46
13:37:32.259806 ARP, Request who-has 10.10.10.90 tell 10.10.10.30, length 46
13:37:32.275694 IP 10.10.10.90.38125 > 10.10.10.30.5900: Flags [S], seq 1933245246, win 1024, options [mss 1460], length 0
```

Conclusion- Detection of Scanning by Snort IDS

When use the -D option, Nmap sends packets to the target from all the specified IP addresses (both decoys and your real IP).

From the target's perspective, it appears to be under a coordinated scan from multiple machines simultaneously. This makes it very difficult for their intrusion detection system (IDS) or firewall to determine which IP is the real attacker.

----- Conclusions -----

During the scanning phase i exercised a broad set of active reconnaissance techniques against the targets within the authorized scope. most scanning technique were not recorded here because the IDS detected a lot of them...

most of the Techniques included are standard Nmap scans (SYN, TCP connect, UDP where applicable), fragmentation and MTU manipulation, packet-length/data-length adjustments, timing and rate tuning, decoy and random-source scans (-D, --rand-source), and off- path/idle (zombie) scans (-sl). Some scan choices were influenced by the need to test Snort's detection behavior (i.e., "IDS-dictated" tests), while others were deliberate attempts to obfuscate scanner origin and evade logging and attribution (source-spoofing and decoy strategies). The results show a split: signature-based rules reliably detected many direct/noisy scans, whereas spoofing/decoy approaches materially reduced attribution and, in several cases, avoided Snort alerts

entirely.

THANK YOU.....